

Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

Supplementary Material

Anonymous submission

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. Public preprocessing produces candidates for 533/548 contexts; the main evaluation retains the full 548-context ground-truth universe and treats the 15 contexts without candidates as empty sets. For sensitivity, the eligible-only denominator contains 3,899 relations, and a documented recovery covers all 548 contexts by lowering the visible-object minimum to two and regenerating views for two scans.

SGFN. We use the official `full_1160` source with its 160-object, 26-relation checkpoint. The evaluated scans, denominator, methods, and tensor-shape checks are recorded in the supplemental package.

Guarantees and Compatibility Diagnostics

Proof of Proposition 1. For any $h \in H_a$, $P_a f(hx) = |H_a|^{-1} \sum_{g \in H_a} f(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $P_a f(hx) = P_a f(x)$. Thus $P_a f$ is constant on each declared orbit. Averaging an already orbit-constant function leaves it unchanged, which proves $P_a^2 = P_a$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton action is the identity.

Proof of Proposition 2. At position j , the routing algorithm pops from the queue indexed by the source family a_j ; hence the output family sequence is exactly $(a_1, \dots, a_n)$. A pass-through queue retains source-relative order, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. For optimality, consider two slots $p < q$ of the same family. If $u(e_p) < u(e_q)$, exchanging them changes the weighted objective by $(\omega_p - \omega_q)(u(e_q) - u(e_p)) \geq 0$. Repeatedly removing such inversions gives the descending family queues used by the route, proving optimality over all family-sequence-preserving permutations. Ties may yield multiple optima without changing either guarantee.

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Counterfactual target construction. Measured-family ground-truth rows provide positives. Negatives are generated within the same scene context and linked to their source positive; no source score, predictor identity, or validation-split row is used. Table 1 gives the complete construction policy.

Table 1: **Counterfactual-negative construction.** Normalized XY distance uses the mean object-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9} .

Family	Intervention and acceptance rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance; require distance ≥ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary; the verifier declares violation only at ≥ 3.5 .
Vertical	On the same directed pair, replace higher than with lower than or conversely; require the positive direction to have $ \Delta z \geq 0.25$ m and normalized $ \Delta z \geq 0.15$.	Shares both signed primitives and thresholds.
Support/contact	Replace the subject or object within context; reject known support/contact pairs in either direction and floor sub-jects; require any two of normalized XY distance ≥ 2.0 , zero overlap, and vertical gap ≥ 0.30 m.	Shares these OBB primitives and thresholds; point-subtype evidence used by the reported verifier is not used to generate negatives.

At most two negatives are emitted per positive, 200 per context–family, and three times the number of positives per family. These caps limit repeated easy negatives; they do not equalize family counts. The export contains 28,977 positives and 37,477 negatives (23,008 proximity, 5,488 vertical, and 37,958 support/contact rows). The explicit overlap above is why verifier-derived V is reported as a related diagnostic rather than independent physical-validity ground truth; the held-out-primitive analysis tests, but does not eliminate, this construct dependence.

RelCompat3D fits one logistic model per measured relation family. Numeric features use training-only means and standard deviations; missing values use the correspond-

ing training mean. Inputs are predicate/family indicators T , predicate-independent pair geometry G , and explicit $T \times G$ interactions. Source score, source rank, source identity, and object class are forbidden. Orbit-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

The 60,208 training rows yield 33,961 linked pairs, and algebra augmentation produces 86,032 optimization rows. We use 800 deterministic full-batch steps at learning rate 0.2; the three family models contain 22–24 parameters. The support/contact model is used only in unrestricted-fusion diagnostics, while the primary family-slot route preserves that family’s source ordering.

Table 2: **Training and development discrimination.** Labels comprise GT-positive/counterfactual pairs, not human physical-validity judgments.

Family	Train	Dev	Param.	Train pairs	AUROC	Brier
Proximity	20,902	2,106	22	9,562	1.0000	.0009
Vertical order	4,922	566	23	1,644	.9982	.0079
Support/contact	34,384	3,574	24	22,755	.9933	.0256

Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.13. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact orbit projection gives maximum proximity-swap and vertical-inverse errors of zero on development and all three final source outputs.

The factor-isolation ablation uses the same split and estimator class. At development, T -only, true raw- G -only, additive $T + G$, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and mismatched-pair geometry reduce the intended reliability signal. These results verify factor use and the implemented relation algebra; they do not convert the constructed target into independent physical-validity ground truth.

Table 3: **Held-out-primitive diagnostic.** Cells are Recall / verifier V under the same family-slot route. Exact removes the normalized scalar consumed by the verifier; family removes all directly related distance or vertical measurements; alternative retains only projected overlap for proximity and horizontal-distance/overlap context for vertical order.

Source	Condition	$K = 50$	$K = 100$
VL-SAT	Source score	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact scalar held out	.9280/.0198	.9660/.0300
	Primitive family held out	.9272/.0268	.9655/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source score	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5692/.0324
	Exact scalar held out	.4383/.0342	.5692/.0325
	Primitive family held out	.4119/.0954	.5257/.0956
	Alternative evidence only	.4106/.1193	.5330/.1188
SGFN	Source score	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact scalar held out	.7465/.0268	.9303/.0357
	Primitive family held out	.7470/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Every held-out model is refitted on the 1,061 training scans with the same orbit-plus-pairwise objective, source-score exclusion, and exact projection. The exact-scalar condition retains development AUROC of .99999 for proximity and .99794 for vertical order. Removing the full primitive family lowers these to .9512 and .7089; alternative evidence yields .8763 and .5531. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full V gain on VL-SAT or SGFN. At $K = 100$, all held-out conditions improve both point metrics across all three predictors; paired scan-cluster intervals are stored for every K . This rules out literal reuse of a single verifier input, not broader construct dependence on correlated geometry.

Table 4: **Unrestricted-fusion mechanism audit at $K = 100$.** Cells are Recall / verifier V. Unlike the primary routed ablation, compatibility is applied to every family; Open3DSG uses the documented full-coverage sensitivity so each corruption has a donor in every context.

Condition	VL-SAT	Open3DSG	SGFN
Structured product	.9688/.0325	.6055/.0339	.9418/.0372
Wrong predicate	.9507/.0745	.5753/.2315	.9182/.1120
Wrong-pair G	.9121/.0525	.2885/.0891	.8429/.0723
Shuffled G	.9089/.0615	.3326/.1418	.8338/.0861
Label-fixed swap	.9509/.0745	.5775/.2325	.9187/.1120
Distance only	.5554/.0981	.5038/.1071	.5554/.0981
Compatibility only	.6186/.0193	.5798/.0321	.6186/.0193

Counterfactual-Policy Sensitivity

We regenerate the training and internal-development targets and refit the proximity and vertical heads under nine one-factor-at-a-time conditions. Each variant recomputes normalization from the 1,061 training scans; no final-validation row

enters target generation, normalization, or fitting. The default refit is bit-exact to the main model. Support/contact remains a source-order pass-through in every condition.

Table 5: **Counterfactual-policy sensitivity.** Order is linked-positive ordering accuracy on the 117-scan internal-development split. Source cells are Recall / verifier V. Only the named factor changes from the default.

(a) $K = 50$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9277/.0197	.4418/.0342	.7450/.0263
Proximity 2.0	1.000	.9280/.0197	.4441/.0342	.7470/.0263
Proximity 3.0	1.000	.9275/.0197	.4423/.0342	.7437/.0264
Vertical 0.20 m	1.000	.9277/.0193	.4408/.0342	.7450/.0258
Vertical 0.30 m	1.000	.9277/.0204	.4439/.0342	.7447/.0274
Negative cap 1	1.000	.9277/.0197	.4431/.0342	.7447/.0263
Negative cap 4	1.000	.9277/.0197	.4418/.0342	.7455/.0263
Pair weight .125	1.000	.9277/.0198	.4421/.0342	.7450/.0264
Pair weight .5	1.000	.9277/.0197	.4418/.0342	.7450/.0262

(b) $K = 100$				
Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9658/.0295	.5692/.0324	.9303/.0350
Proximity 2.0	1.000	.9660/.0294	.5733/.0324	.9325/.0350
Proximity 3.0	1.000	.9653/.0296	.5662/.0324	.9277/.0350
Vertical 0.20 m	1.000	.9658/.0289	.5687/.0324	.9308/.0334
Vertical 0.30 m	1.000	.9658/.0306	.5680/.0324	.9300/.0370
Negative cap 1	1.000	.9658/.0295	.5697/.0324	.9293/.0350
Negative cap 4	1.000	.9658/.0295	.5680/.0324	.9305/.0350
Pair weight .125	1.000	.9658/.0296	.5695/.0324	.9303/.0351
Pair weight .5	1.000	.9658/.0294	.5687/.0324	.9303/.0347

Across all variants, the maximum absolute change from the default is .0023 Recall and .0011 V at $K = 50$, and .0040 Recall and .0020 V at $K = 100$. Every variant preserves the source-score point-estimate improvement on both metrics for all three predictors at $K = 50$ and $K = 100$. This reduces dependence on a particular threshold or negative count, but does not make the constructed target independent of the verifier.

Strong-Comparator and Open3DSG Sensitivities

Same-route strong comparators. To separate score construction and capacity from routing, rank-average, RRF, and a small two-hidden-unit MLP use the same family-slot sequence, support/contact pass-through, public/full 548-context target, and scan-cluster resampling as the linear product. The MLP uses the same training rows, constructed target, feature contract, source-score exclusion, and split as the linear heads; it is fitted once and applied to all three predictors without target-specific refitting.

Table 6: **Same-route fusion comparisons.** Cells are Recall / verifier V at $K = 50$ and $K = 100$. Every row preserves the source family sequence and support/contact selections.

Source	Fusion	$K = 50$	$K = 100$
VL-SAT	Product	.9277/.0197	.9658/.0295
	Matched MLP	.9272/.0189	.9650/.0296
	Rank-average	.9026/.0162	.9700/.0225
	RRF	.9174/.0180	.9710/.0247
Open3DSG	Product	.4418/.0342	.5692/.0324
	Matched MLP	.4670/.0413	.5989/.0371
	Rank-average	.4323/.0458	.5408/.0548
	RRF	.4237/.0948	.5476/.0772
SGFN	Product	.7450/.0263	.9303/.0350
	Matched MLP	.7457/.0258	.9288/.0350
	Rank-average	.7022/.0236	.9280/.0259
	RRF	.6813/.0253	.8774/.0302

At $K = 50$, the matched MLP has higher Open3DSG Recall than the product but also higher V; product and MLP are nearly tied on VL-SAT and SGFN. Rank fusion can lower V at large budgets but incurs larger predictor-dependent Recall losses. No fusion dominates the joint trade-off. A separate exact-label MLP uses stronger SGFN-specific correctness supervision and is therefore retained only as an unrestricted upper comparator, not as an apples-to-apples baseline.

Family-conditioning ablation. The pooled product fits one compatibility model across all families and does not use the primary family-slot route. Compared with the unrestricted family-specific product, it can increase Recall but generally raises V, particularly for Open3DSG and SGFN:

Table 7: **Family-specific versus pooled unrestricted products.** Cells are Recall / verifier V at $K = 50$ and $K = 100$.

Source	$K = 50$		$K = 100$	
	Family	Pooled	Family	Pooled
VL-SAT	.9293/.0203	.9300/.0219	.9688/.0325	.9690/.0387
Open3DSG	.4655/.0265	.4675/.0510	.6005/.0330	.6402/.0743
SGFN	.7706/.0256	.7925/.0292	.9418/.0372	.9413/.0464

Open3DSG public-route sensitivity. The public preprocessing drops a context when fewer than four annotated objects retain view metadata. This accounts for all 15 missing contexts. Table 8 distinguishes the faithful eligible-context evaluation, the conservative full-target main evaluation, and a documented full-coverage recovery.

Table 8: **Open3DSG route sensitivity at $K = 100$.** Cells are Recall / verifier V. The main route uses the full 3,972-relation denominator and no predictions for the 15 public-pipeline drops.

Route	Contexts / GT	Source	Routed RelCompat3D
Public eligible	533 / 3,899	.5206/.1242	.5799/.0324
Public, full target (main)	548 / 3,972	.5111/.1242	.5692/.0324
Recovered full coverage	548 / 3,972	.5161/.1242	.5743/.0332

The conclusion is stable to all three denominator and coverage choices. The main paper uses the conservative full-target public route; recovered results are sensitivity evidence rather than the headline benchmark.

Scan-Cluster Confidence Intervals

The analysis resamples the 157 scan identifiers 1,000 times and carries every context of a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions. A paired 548-context bootstrap gives the same directions and is retained as a finer-grained sensitivity.

Table 9: **Scan-cluster intervals at $K = 50$.** Entries are routed RelCompat3D minus source score with paired 95% intervals.

Source	Δ Recall	Δ Verifier-V
VL-SAT	+0.005 [−.0005,.0019]	−.0070 [−.0088,−.0055]
Open3DSG	+0.0375 [.0296,.0449]	−.1045 [−.1105,−.0986]
SGFN	+0.0048 [.0028,.0070]	−.0122 [−.0142,−.0102]

At $K = 50$, every Verifier-V interval remains below zero. Recall improves on Open3DSG and SGFN, while the near-saturated VL-SAT interval is centered near zero. At $K = 100$, the changes have the same direction and strictly positive Recall intervals for all three predictors; its point estimates remain in the main five-budget table.

External Transfer Stress Test

We apply the 3DSSG-trained compatibility model and family-slot rule without target-specific refitting to the official ReplicaSSG test scenes using FROSS predictions. No ReplicaSSG row is used for fitting, normalization, imputation, or hyperparameter selection. The exact mapping covers near, above, and under; support/contact is excluded because the ontologies do not provide an exact mapping. We report this as a transfer stress test rather than evidence of dataset-level generalization.

Table 10: **ReplicaSSG/FROSS transfer across all reported budgets.** R is exact-label Recall over 172 relations and V is verifier-derived Violation (lower is better). The route uses the same model and ranking rule as the main experiment.

K	Source R/V	RelCompat3D R/V	Δ R/ Δ V
5	.0465/.0545	.0698/.0182	+0.0233/−.0364
10	.0756/.0818	.1453/.0273	+0.0698/−.0545
20	.1453/.1273	.2209/.0364	+0.0756/−.0909
50	.2616/.1328	.3140/.0904	+0.0523/−.0424
100	.3547/.1967	.3547/.1958	+0.0000/−.0010

Paired scene-bootstrap intervals support joint improvement at $K = 10$ (Δ R [.0048,.1467], Δ V [−.1000,−.0091]) and $K = 50$ (Δ R [.0179,.0811], Δ V [−.0683,−.0185]). The $K = 20$ Recall interval touches zero, while $K = 5$ is inconclusive. At $K = 100$, the product changes only one selected row: 86.28% of source scores are exactly zero, and unresolved boundary ties in 9 of 11 scenes make multiplicative re-ranking nearly inert. Candidate support also caps exact-label Recall at 76/172 (44.19%). The stress test therefore supports budget-dependent transfer, not dataset-level generalization.

Family Composition at the Global $K = 100$ Operating Point

Table 11 reports each family’s slice inside the actual global top-100 ranking. Deltas compare routed RelCompat3D with the source score using shared paired context-bootstrap indices.

Table 11: **Global-top-100 family slices.** Positive Δ V is a regression. Values are verifier-derived.

Source	Family	Δ R	Δ V
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+0.0040	−.0076
VL-SAT	Vertical order	+0.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+0.0968	−.0443
Open3DSG	Vertical order	+0.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+0.0142	−.0150
SGFN	Vertical order	+0.0051	−.0643

Support/contact is exactly unchanged because the route preserves both its selection and the source-ranked family composition at every K . Vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. Full within-family and global-slice results for $K \in \{5, 10, 20, 50, 100\}$, including simultaneous family-wise intervals, are included in the supplemental package.

Relative-Size Scope Extension

We evaluate bigger than/smaller than as secondary breadth evidence, not as core evidence that learned

compatibility is necessary. The extension uses the same 1,061/117/157 train/development/evaluation split. Its model receives predicate sign, robust point extents and volume ratios, and signed predicate–geometry interactions; source score, rank, object class, and evaluation labels are excluded. Alternating point subset A (p05–p95 extents) provides model evidence, whereas disjoint subset B (p02–p98 extents) provides the verifier measurement. Annotation OBB volume is used only by a fixed-rule baseline.

The model fits 3,244 training rows and obtains target-discrimination AUROC .9974 on 376 development rows. Correct-predicate ordering is 96.81%; exact inverse-equivariance and common-scale invariance have maximum error 2.22×10^{-16} . Table 12 summarizes the $K = 100$ result; the supplemental package contains all five K values and family compositions.

Table 12: **Relative-size extension at $K = 100$.** Within-size Recall is unchanged for all sources. Global columns use the four-family denominator of 4,142; parentheses are paired 95% context-bootstrap intervals.

Source	Within-size ΔV	Global ΔR	Global ΔV
VL-SAT	−.1025 [−.1063, −.0988]	+0.048 [.0020, .0080]	−.0239 [−.0261, −.0218]
Open3DSG	−.1274 [−.1383, −.1167]	+0.0698 [.0478, .0935]	−.1040 [−.1090, −.0997]
SGFN	−.0593 [−.0616, −.0572]	+0.0256 [.0199, .0317]	−.0359 [−.0381, −.0336]

The fixed robust-point rule is as strong or stronger than the learned model on within-family violation: learned-minus-rule ΔV is +.00318 for VL-SAT, +.00005 for Open3DSG, and +.00367 for SGFN. Disjoint point subsets remove exact rule reuse but not shared-geometry construct dependence. The family therefore broadens the measured scope while remaining a rule-easy secondary analysis.

Verifier-Uncertainty Sensitivity

The reported $V@K_{all}$ includes uncertain rows in its denominator but does not label them satisfied. We therefore recompute each ranking with $V@K_{dec}$ over satisfied/violated rows, uncertainty rate $U@K$, and a pessimistic bound $V@K_{pess}$ that counts every uncertain row as a violation. Table 13 gives the $K = 100$ comparison.

Table 13: **Uncertainty sensitivity at $K = 100$.** Lower is better for all four reported quantities.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	RelCompat3D routed	.0295	.0452	.3473	.3768
Open3DSG	Source score	.1242	.2126	.4161	.5402
Open3DSG	RelCompat3D routed	.0324	.0548	.4084	.4408
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	RelCompat3D routed	.0350	.0565	.3803	.4153

Routed-minus-source changes in V_{dec} are −.0277 for VL-SAT, −.1578 for Open3DSG, and −.0440 for SGFN;

their paired 95% intervals all exclude zero. Uncertainty-rate changes are +.0009, −.0077, and +.0071, whereas pessimistic- V changes are −.0173, −.0994, and −.0209. Thus the reported conclusion holds without requiring uncertainty to decrease uniformly. The hard-rule baseline has zero all-status V but retains uncertain rows and therefore has pessimistic V of .3712/.4722/.4042. Thus zero measured violation does not imply reliable or complete selection.